Curriculum and Instruction

UNIVERSITY OF MINNESOTA
Driven to Discover®

M. Ed. / Initial Licensure in Science Education: Grades 9-12 Physics
Prerequisite Course Evaluation

Candidates are required to have an undergraduate degree from an accredited institution with a major in Physics **or** closely related field, or meet the prerequisites below, and with appropriate liberal arts/general education coursework. Courses in each of the content areas are required to meet Standard 1 of MNSEPT: "*The learner understands and can create learning experiences that make these aspects of content matter meaningful for students" (https://www.revisor.mn.gov/rules/?id=8710.2000).*

Candidate Name:	**UMN ID#:**	**Email:**	**Date:**
Degree Granting Institution:	**Major(s):**	**Minor(s):**	**Prepared by:**

Physics Major Status	***Major***	**Degree Granting Institution/ Degree Year**	**Physics Degree Requirement Check ✔**
• An undergraduate physics degree from an accredited institution in the U.S. or an equivalent 4 year physics undergraduate degree from an accredited international institution.			

CANDIDATES WITHOUT A PHYSICS MAJOR MUST MEET COURES FOR AN EQUIVALENT OF AN UNDERGRADUATE MAJOR. Per U of MN program requirements, 39 credits are required for a Physics degree (including General Psychology). Adjustment can be made for Non U of M degree programs with varying credit requirements.

Content Area	**Requirement**	**U of M Course Options**	**Course/Term/Year**	**Credits /Grade**	**Institution If Other Than Degree Granting**
Psychology	General Psychology (4 credits)	PSY 1001			
Introductory Physics (three courses)	Physics I (4 credits)	PHYS 1101, 1221, or 1301W			
	Physics II (4 credits)	PHYS 1102, 1202W, or 1302W			
	Physics III (4 credits)	PHYS 2303 or 2503			
Quantum Physics (one course, with lab)	Quantum Physics (4 credits)	PHYS 2601			
	Quantum Physics Lab (3 credits)	PHYS 3605W			
Upper division topical physics (choose two courses)	Analytical Mechanics (4 credits)	PHYS 4001			
	Electricity and Magnetism (4 credits)	PHYS 4002			

	Quantum Mechanics (4 credits)	PHYS 4101			
	Statistical and Thermal Physics (3 credits)	PHYS 4201			
Directed Research in Physics (two courses)	Experimental Physics I (5 credits)	PHYS 4051			
	Experimental Physics II (5 credits)	PHYS 4052W			

Additional Licensure in Grades 5-8 General Science (Optional)
Prerequisite Course Evaluation

Content Area	Requirement	U of M Course Options	Course/Term/Year	Credits/Grade	Institution
Life Science two courses (with labs)	Biology I	BIOL 1001 or 1009 or 1951/1961			
	Biology II	BIOL 2003/3004 or 1055 or GCD 3022 or 3033			
Physics sequence two courses (with labs)	Physics I	PHYS 1101 or 1221 or 1301W			
	Physics II	PHYS 1102 or 1222 or 1302W			
Chemistry sequence two courses (with labs)	General Chemistry I	CHEM 1061 & 1065 or 1081 & 1065 or 1071H & 1075H			
	General Chemistry II	CHEM 1062 & 1066 or 1082 & 1086 or 1072H & 1076H			
Earth science two courses (with labs)	Physical Geology	ESCI 1001			
	Astronomy	AST 1001			